

Doc 4: Vectors, Dot Product, Norms

Module 1: Linear algebra | Theory recap (read first, about 10 minutes)

1. The dot product: multiply matching entries, then add

Both vectors must have the same length. The answer is a **single number**, not a vector. You already did this in Doc 2 as a weighted sum.

Formula $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + \dots + a_nb_n = \sum_{i=1}^n a_ib_i$

Example $(1, 2, 3) \cdot (4, 5, 6)$ $1 \times 4 + 2 \times 5 + 3 \times 6 = 4 + 10 + 18 = 32$

What the sign tells you positive: pointing the same general way zero: perpendicular. negative: opposite way

2. Length (norm) and distance

The **length** of a vector, called its **norm**, comes from the Pythagorean theorem: square each entry, add, take the square root. Example: (3, 4) has length $\sqrt{9 + 16} = \sqrt{25} = 5$.

L2 norm (usual length) $\|\mathbf{a}\| = \sqrt{a_1^2 + \dots + a_n^2} = \sqrt{\mathbf{a} \cdot \mathbf{a}}$ (2, 3, 6): $\sqrt{4 + 9 + 36} = 7$

L1 norm $\|\mathbf{a}\|_1 = |a_1| + |a_2| + \dots + |a_n|$ (1, -2, 2): $1 + 2 + 2 = 5$

Unit vector $\hat{\mathbf{a}} = \mathbf{a} / \|\mathbf{a}\|$ (3, 4) / 5 = (0.6, 0.8), length 1

Distance $d(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|$ (1, 5) and (4, 1): (-3, 4), so $d = 5$

3. Cosine similarity: do two vectors point the same way?

Formula $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$ always between -1 and 1

Reading it 1 = same direction, 0 = perpendicular -1 = exactly opposite

Why it works $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$ cosine = dot product with the lengths divided out

4. Where machine learning uses these

- **A neuron** computes a dot product of weights and inputs, then adds a bias.
- **Attention** (Doc 29) scores how well two words match with dot products.
- **Cosine similarity** compares embeddings: are two words or documents about the same thing?
- **Norms** measure error size and penalize large weights (regularization). **Distance** powers nearest-neighbor search.

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Worked examples: every cell filled in. Read them, then cover the answers and redo them.

Example 1: Dot product on a grid (with a negative)

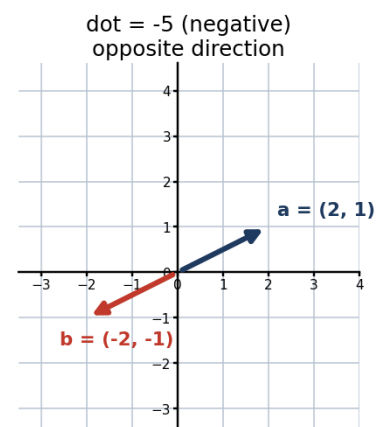
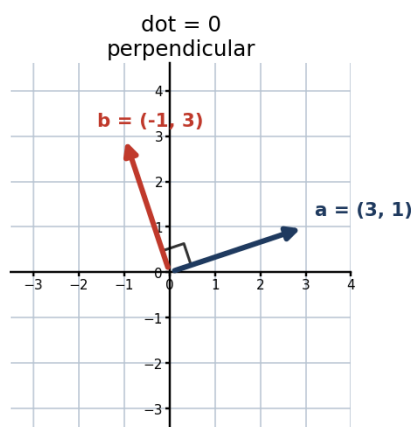
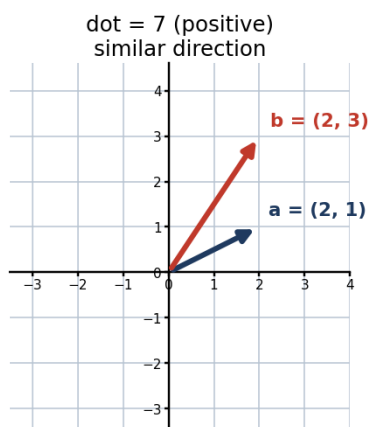
i	1	2	3
a_i	2	-1	3
b_i	1	4	2
$a_i \times b_i$	2	-4	6
Dot product	$2 + (-4) + 6 = 4$		

Example 2: Norm on a grid

i	1	2	3
a_i	1	2	2
a_i^2	1	4	4
Sum of squares	$1 + 4 + 4 = 9$		
Norm = square root	$\text{sqrt}(9) = 3$		

Two entries work the same way: (3, 4) gives $9 + 16 = 25$, and the norm is $\text{sqrt}(25) = 5$.

Example 3: What the dot product tells you



Left: $(2,1) \cdot (2,3) = 4 + 3 = 7$. Middle: $(3,1) \cdot (-1,3) = -3 + 3 = 0$. Right: $(2,1) \cdot (-2,-1) = -4 - 1 = -5$.

Example 4: Cosine similarity

Vectors	$a = (3, 4)$ $b = (4, 3)$
Dot product $a \cdot b$	$12 + 12 = 24$
Norm of a	$\text{sqrt}(9 + 16) = 5$
Norm of b	$\text{sqrt}(16 + 9) = 5$
Cosine	$\text{dot} / (\text{norm } a \times \text{norm } b) = 24 / (5 \times 5) = 0.96$

0.96 is close to 1, so a and b point in almost the same direction.

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Exercises: use a pencil. Yellow cells are yours to fill in. Name: _____ Date: _____

Cheat sheet

n	1	2	3	4	5	6	7	8	9	10	11	12	13
n squared	1	4	9	16	25	36	49	64	81	100	121	144	169
Roots: $\sqrt{2} = 1.414$, $\sqrt{5} = 2.236$, $\sqrt{20} = 4.472$													

A. Dot products

i	1	2	3	4
a_i	3	-2	1	4
b_i	2	5	-3	1
$a_i \times b_i$				
Dot product				

i	1	2	3
u_i	1	0	2
v_i	0	7	0
$u_i \times v_i$			
Dot product			

Are u and v perpendicular? _____

B. Norms

Vector	Sum of squares	L2 norm (length)	L1 norm (sum of entries)
(6, 8)			
(5, 12)			
(2, 3, 6)			
(1, -2, 2)			

C. Unit vectors

Vector a	Norm of a	Unit vector = a / norm	Length of the unit vector
(3, 4)			
(1, 2, 2)			

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Exercises, continued

D. Cosine similarity

Pair	Dot product	Norm of first	Norm of second	Cosine
1. $\mathbf{a} = (1, 2, 2)$ $\mathbf{b} = (2, 3, 6)$				
2. $\mathbf{u} = (2, 1)$ $\mathbf{v} = (-1, 2)$				
3. $\mathbf{a} = (1, 2)$ $\mathbf{c} = (-2, -4)$				

Perpendicular pair: _____ Exactly opposite pair: _____

E. Nearest neighbor: which house is most like the query?

Features are (size, bedrooms, age). The query house is $\mathbf{x} = (2, 3, 1)$. For each candidate, find the difference $\mathbf{x} - \text{candidate}$, the sum of its squares, and the distance.

Candidate	Difference $\mathbf{x} - \text{candidate}$	Sum of squares	Distance
$\mathbf{H1} = (4, 1, 0)$			
$\mathbf{H2} = (2, 3, 3)$			
$\mathbf{H3} = (5, 7, 1)$			

The nearest house is _____.

F. Challenge: one neuron

A neuron has weights $\mathbf{w} = (2, -1, 3)$ and bias $\mathbf{b} = 1$. Compute $z = \mathbf{w} \cdot \mathbf{x} + \mathbf{b}$, then apply ReLU (from Doc 2): $\text{ReLU}(z) = \max(0, z)$.

Input $\mathbf{x}$	$\mathbf{w} \cdot \mathbf{x}$	$z = \mathbf{w} \cdot \mathbf{x} + \mathbf{b}$	$\text{ReLU}(z)$
$\mathbf{x} = (1, 4, 2)$			
$\mathbf{x} = (0, 5, 1)$			

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Answer key: check your work only after finishing

A. Dot products

i	1	2	3	4
$a_i \times b_i$	6	-10	-3	4
Dot product	$6 - 10 - 3 + 4 = -3$			
i	1	2	3	
$u_i \times v_i$	0	0	0	
Dot product	0			

Yes: $u \cdot v = 0$, so u and v are perpendicular.

B and C. Norms and unit vectors

Vector	Sum of squares	L2 norm	L1 norm
(6, 8)	100	10	14
(5, 12)	169	13	17
(2, 3, 6)	49	7	11
(1, -2, 2)	9	3	5

Vector a	Norm	Unit vector	Its length
(3, 4)	5	(0.6, 0.8)	1
(1, 2, 2)	3	(1/3, 2/3, 2/3)	1

Check: $0.36 + 0.64 = 1$ and $1/9 + 4/9 + 4/9 = 1$.

D. Cosine similarity

Pair	Dot	Norm 1	Norm 2	Cosine
1. $a = (1, 2, 2)$ $b = (2, 3, 6)$	20	3	7	$20/21 = 0.952$
2. $u = (2, 1)$ $v = (-1, 2)$	0	2.236	2.236	0
3. $a = (1, 2)$ $c = (-2, -4)$	-10	2.236	4.472	-1

Perpendicular: pair 2 (cosine 0). Exactly opposite: pair 3 (cosine -1).

E and F

Candidate	Difference	Sum of squares	Distance
H1 = (4, 1, 0)	(-2, 2, 1)	9	3
H2 = (2, 3, 3)	(0, 0, -2)	4	2
H3 = (5, 7, 1)	(-3, -4, 0)	25	5

Nearest house: H2 (distance 2).

Input x	$w \cdot x$	z	ReLU(z)
$x = (1, 4, 2)$	4	5	5
$x = (0, 5, 1)$	-2	-1	0

$w \cdot x = 2 \times 1 - 1 \times 4 + 3 \times 2 = 4$ for the first input, and $0 - 5 + 3 = -2$ for the second.

Key takeaways: dot product = multiply matching entries and add; norm = square root of the dot product with itself; cosine = dot / (norm $\times$ norm); distance = norm of the difference; a neuron is a dot product plus a bias.

Next: Doc 5 - Matrix multiplication.